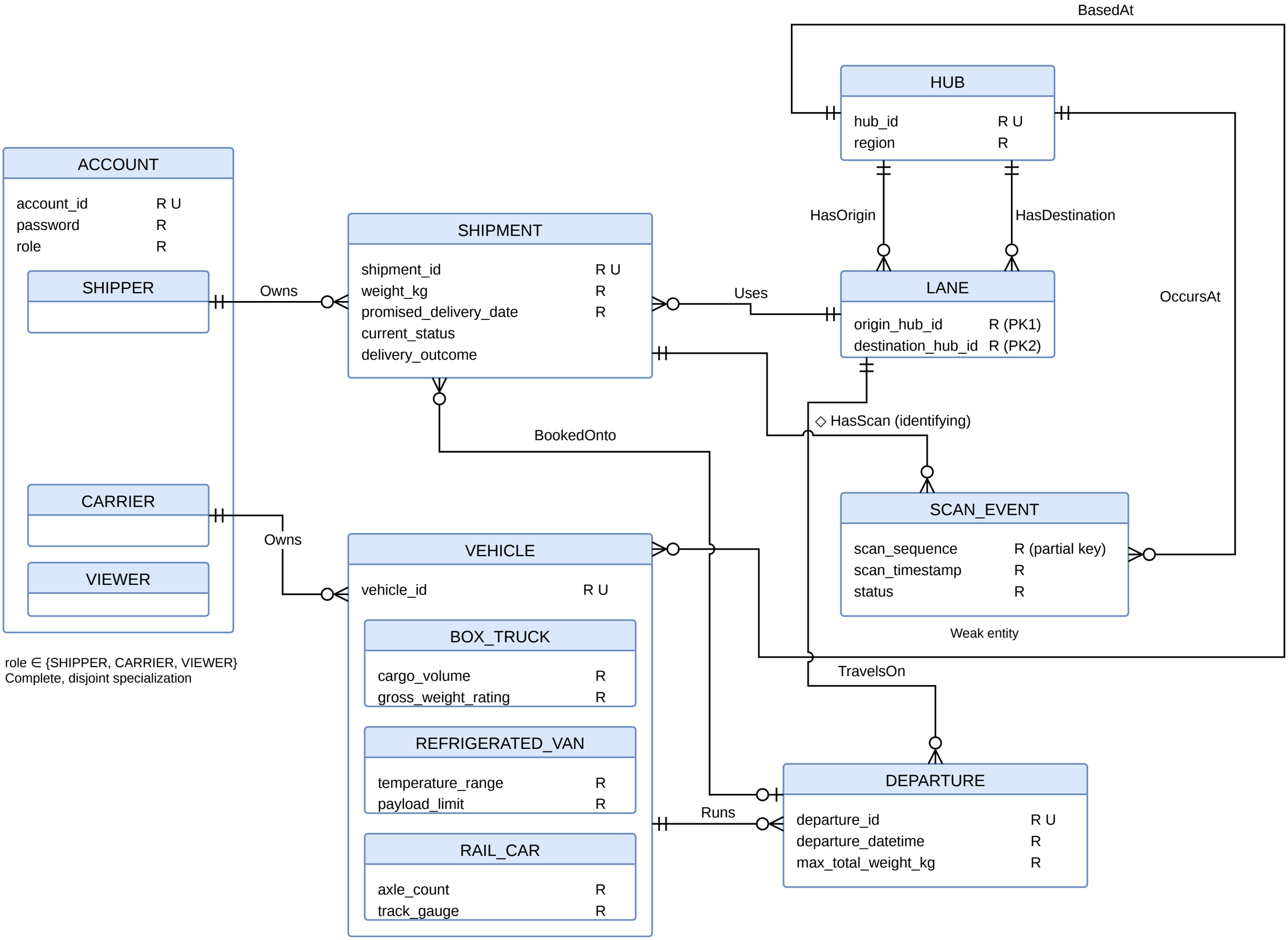

BUSINESS RULES

- HUB.region ∈ {Northeast, Southeast, Midwest, Southwest, West}.
- The LANE key is (origin_hub_id, destination_hub_id). The hubs must differ.
- HUB and LANE are preloaded reference data. Shipments and departures must use an existing valid directed lane.
- SHIPMENT.weight_kg > 0 and DEPARTURE.max_total_weight_kg > 0.
- BookedOnto assumes at most one departure per shipment. Each departure may carry many shipments.
- The booking procedure atomically checks matching lanes and available capacity using the sum of booked shipment weights.
- SCAN_EVENT is a weak entity identified through SHIPMENT; its key is (shipment_id, scan_sequence).
- SCAN_EVENT.status and SHIPMENT.current_status use {RECEIVED, IN_TRANSIT, ARRIVED_AT_HUB, OUT_FOR_DELIVERY, DELIVERED}.
- The scan chronology trigger enforces valid transitions and nondecreasing timestamps. Intermediate-hub movement may alternate between IN_TRANSIT and ARRIVED_AT_HUB.
- SHIPMENT.delivery_outcome ∈ {ON_TIME, LATE, NULL}; it remains NULL until delivery.
- Scan insertion triggers maintain current_status and determine delivery_outcome from the promised delivery date.